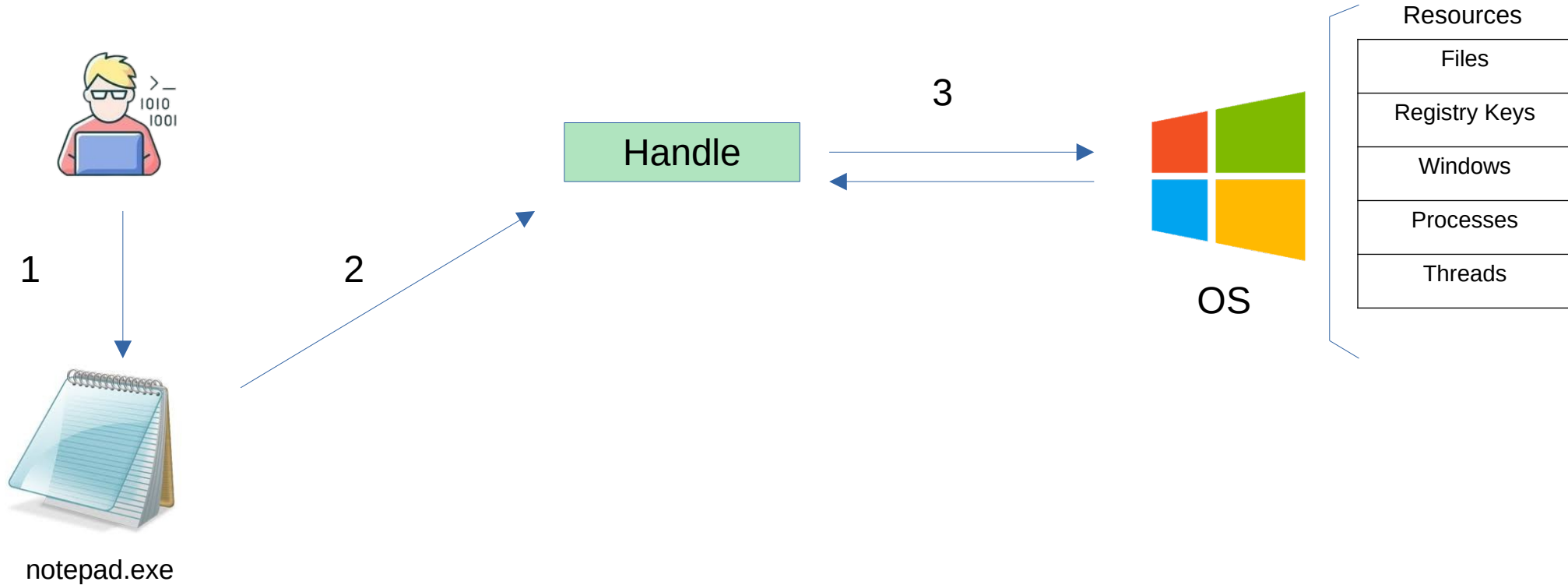


Understanding Handle

Handles: When you request something from Windows, it gives you a handle so that it can track your request. This handle helps Windows identify and manage different resources like files, windows, and processes.



Handles: When you request something from Windows, it gives you a handle so that it can track your request. This handle helps Windows identify and manage different resources like files, windows, and processes.

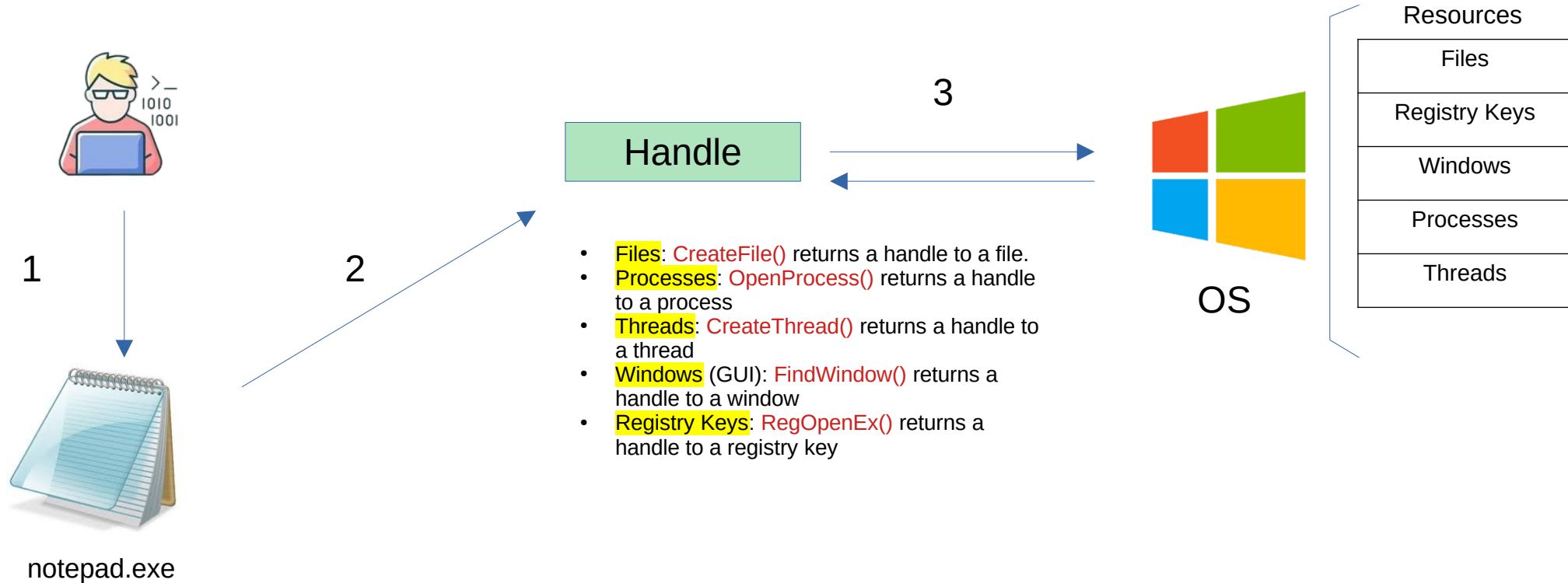


Table of Windows Handle Types

Handle Type	Data Type	Represents	Used With	Example Function to Get Handle	Notes
Window Handle	HWND	A window created by an application	<code>FindWindow()</code> , <code>GetForegroundWindow()</code> , <code>ShowWindow()</code>	<code>FindWindow(class, title)</code>	Used to manipulate windows (resize, move, etc.)
Process Handle	HANDLE	A running process	<code>OpenProcess()</code> , <code>GetCurrentProcess()</code> , <code>TerminateProcess()</code>	<code>OpenProcess(AccessRights, FALSE, PID)</code>	Needed for process manipulation (memory reading, injection)
Thread Handle	HANDLE	A thread within a process	<code>OpenThread()</code> , <code>CreateThread()</code> , <code>SuspendThread()</code>	<code>OpenThread(AccessRights, FALSE, TID)</code>	Allows control over specific threads in a process
File Handle	HANDLE	An open file	<code>CreateFile()</code> , <code>ReadFile()</code> , <code>WriteFile()</code>	<code>CreateFile(path, Access, Share, NULL, OPEN_EXISTING, 0, NULL)</code>	Required for file I/O operations
Registry Handle	HKEY	A registry key	<code>RegOpenKeyEx()</code> , <code>RegQueryValueEx()</code> , <code>RegSetValueEx()</code>	<code>RegOpenKeyEx(HKEY_LOCAL_MACHINE, path, 0, Access, &hKey)</code>	Used to read/write Windows registry values
Event Handle	HANDLE	A synchronization event	<code>CreateEvent()</code> , <code>SetEvent()</code> , <code>ResetEvent()</code>	<code>CreateEvent(NULL, TRUE, FALSE, L"EventName")</code>	Used to signal between threads/processes
Mutex Handle	HANDLE	A mutual exclusion object	<code>CreateMutex()</code> , <code>ReleaseMutex()</code>	<code>CreateMutex(NULL, FALSE, L"MutexName")</code>	Ensures only one thread accesses a resource at a time

Semaphore Handle	HANDLE	A counting semaphore	CreateSemaphore(), ReleaseSemaphore()	CreateSemaphore(NULL, INITIAL_COUNT, MAX_COUNT, L"SemName")	Controls multiple access to a resource
Pipe Handle	HANDLE	A named or anonymous pipe	CreatePipe(), CreateNamedPipe(), ReadFile()	CreateNamedPipe(name, mode, pipeType, ...)	Used for inter-process communication (IPC)
Token Handle	HANDLE	A security token	OpenProcessToken(), DuplicateTokenEx()	OpenProcessToken(hProcess, ACCESS, &hToken)	Stores user security info (privileges, groups)
Job Handle	HANDLE	A job object (group of processes)	CreateJobObject(), AssignProcessToJobObject()	CreateJobObject(NULL, L"JobName")	Allows managing multiple processes together
Console Handle	HANDLE	A console input/output	GetStdHandle(), WriteConsole()	GetStdHandle(STD_OUTPUT_HANDLE)	Used for reading/writing to the console
Device Handle	HANDLE	A device driver interface	CreateFile() [for \\.\DeviceName]	CreateFile(L"\\\\.\\PhysicalDrive0", ...)	Accesses hardware like disks, serial ports
Service Handle	SC_HANDLE	A Windows service	OpenService(), StartService()	OpenService(hSCManager, L"ServiceName", ACCESS)	Used to start, stop, configure Windows services
Desktop Handle	HDESK	A desktop object	OpenDesktop(), SwitchDesktop()	OpenDesktop(L"Winlogon", 0, FALSE, ACCESS)	Manages GUI desktops (like login screen)
Station Handle	HWINSTA	A window station	OpenWindowStation(), SetProcessWindowStation()	OpenWindowStation(L"WinSta0", FALSE, ACCESS)	A collection of desktops, used for GUI isolation

Timer Handle	HANDLE	A waitable timer object	CreateWaitableTimer(), SetWaitableTimer()	CreateWaitableTimer(NULL, FALSE, L"TimerName")	Used for precise timing operations
Memory Mapping Handle	HANDLE	A shared memory object	CreateFileMapping(), MapViewOfFile()	CreateFileMapping(INVALID_HANDLE_VALUE, ...)	Allows different processes to share memory
Clipboard Handle	HANDLE	Data stored in the clipboard	OpenClipboard(), GetClipboardData()	OpenClipboard(NULL)	Allows access to clipboard contents
Accelerator Handle	HACCEL	Keyboard shortcut table	CreateAcceleratorTable(), TranslateAccelerator()	CreateAcceleratorTable(accelArray, numItems)	Maps keyboard shortcuts to commands
Menu Handle	HMENU	A menu of a window	CreateMenu(), GetMenu(), SetMenu()	GetMenu(hWnd)	Manages application menus
Cursor Handle	HCURSOR	A mouse cursor	LoadCursor(), SetCursor()	LoadCursor(NULL, IDC_ARROW)	Changes the cursor appearance
Icon Handle	HICON	An application icon	LoadIcon(), DrawIcon()	LoadIcon(NULL, IDI_APPLICATION)	Represents an icon used in an application

Handle Example: Find out running notepad program

Handle Type	Data Type	Represents	Used With	Example Function to Get Handle	Notes
Window Handle	HWND	A window created by an application	<code>FindWindow()</code> , <code>GetForegroundWindow()</code> , <code>ShowWindow()</code>	<code>FindWindow(class, title)</code>	Used to manipulate windows (resize, move, etc.)

Handle

```
#include<iostream>
#include<Windows.h>

int main()
{
    HWND hWnd = FindWindow(NULL,"Untitled - Notepad");

    if(hWnd) {
        std::cout << "Notepad window found" << std::endl;
    }
    else
    {
        std::cout << "Notepad window not found" << std::endl;
    }
    return 0;
}
```